

Exercises on the geometry of linear equations

Problem 1.1: (1.3 #4. *Introduction to Linear Algebra*: Strang) Find a combination $x_1 \mathbf{w}_1 + x_2 \mathbf{w}_2 + x_3 \mathbf{w}_3$ that gives the zero vector:

$$\mathbf{w}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \quad \mathbf{w}_2 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} \quad \mathbf{w}_3 = \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix}.$$

There is a combination of the vectors that gives the zero vector.

Those vectors are (independent)(dependent). ✓

$$x_1 = x_3 = -\frac{1}{2} x_2$$

The three vectors lie in a _____. The matrix W with those columns is not invertible.

3-d ~~space~~ plane

Problem 1.2: Multiply: $\begin{bmatrix} 1 & 2 & 0 \\ 2 & 0 & 3 \\ 4 & 1 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 9 \\ 11 \end{bmatrix}$ ✓

Problem 1.3: True or false: A 3 by 2 matrix A times a 2 by 3 matrix B equals a 3 by 3 matrix AB . If this is false, write a similar sentence which is correct.

True ✓

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